

Lab 11: Cities

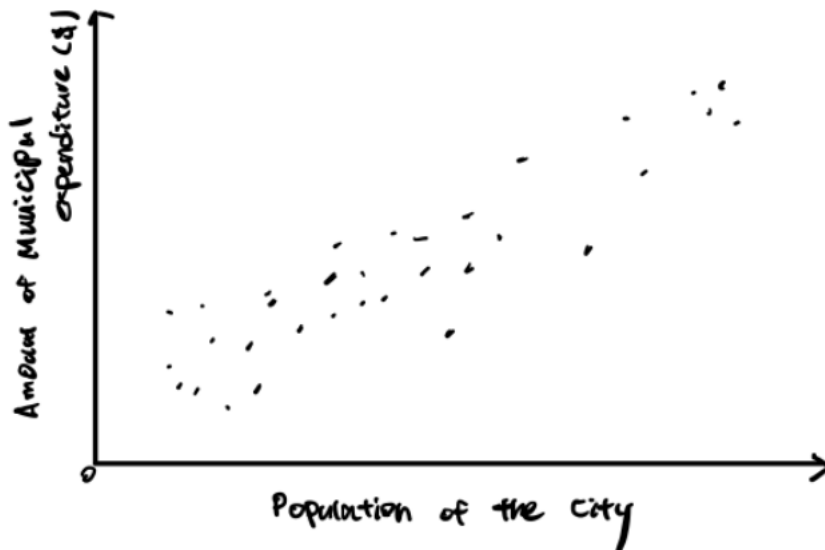
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1. What do you think the distribution of the variable pop looks like? Which plot would you use to show this distribution?

The distribution of the variable pop could look right skewed, because there is more smaller to medium sized cities in Brazil than super large ones. We could use a density plot or a histogram to show this distribution.

2. How would you plot these two variables (Sketch)? Describe how you expect the relation to look like. Is it linear? If yes, is it a positive or negative relation?



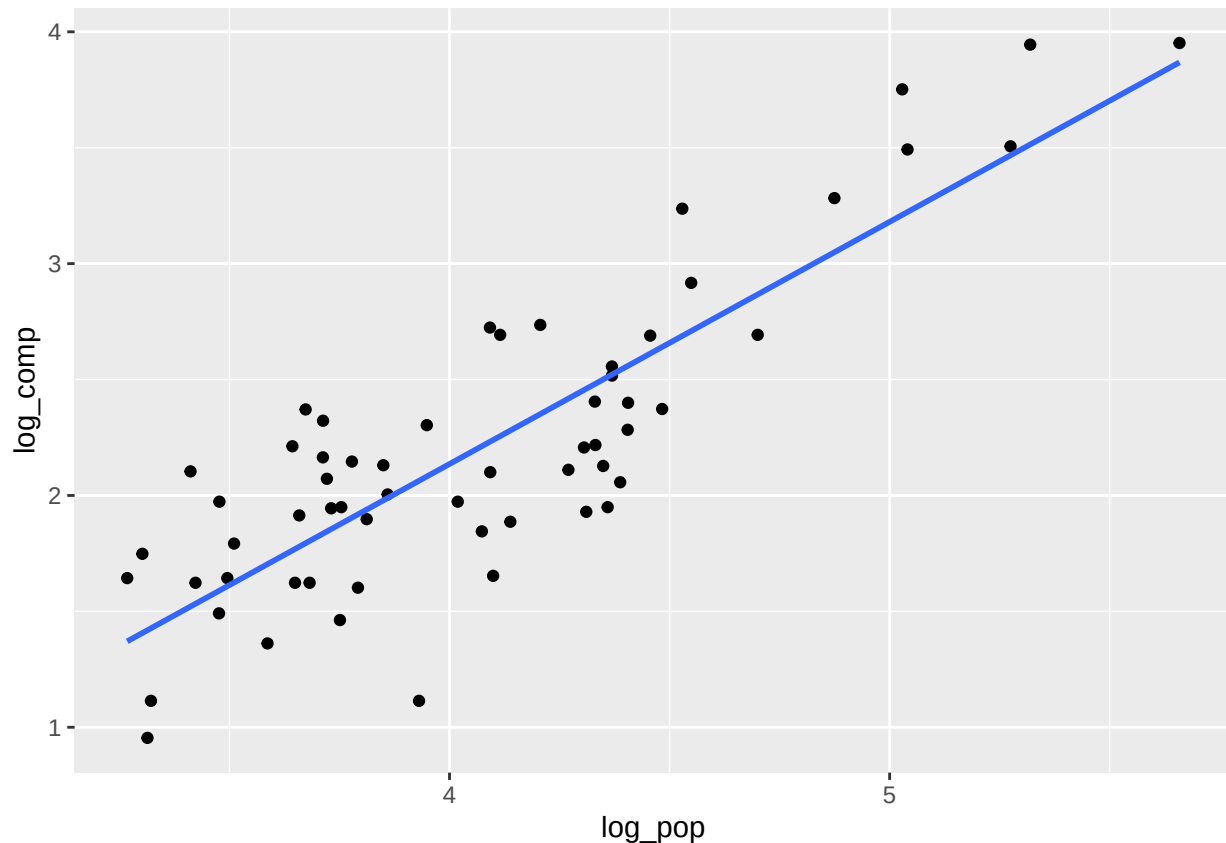
As the population increase, the municipal expenditures will also increase. This could be a moderate to strong linear and positive relationship.

3. Create two new columns: the log (base 10) of population and the number of companies and use them to construct a scatterplot. How would you describe the relation between these two variables? What is its form, strength and direction?

```
cit_log<-cit %>%
  mutate(log_pop = log10(pop),
         log_comp = log10(comp))

ggplot(cit_log, aes (x = log_pop, y = log_comp)) + geom_point() + stat_smooth(method = "lm", se = F)

## 'geom_smooth()' using formula 'y ~ x'
```



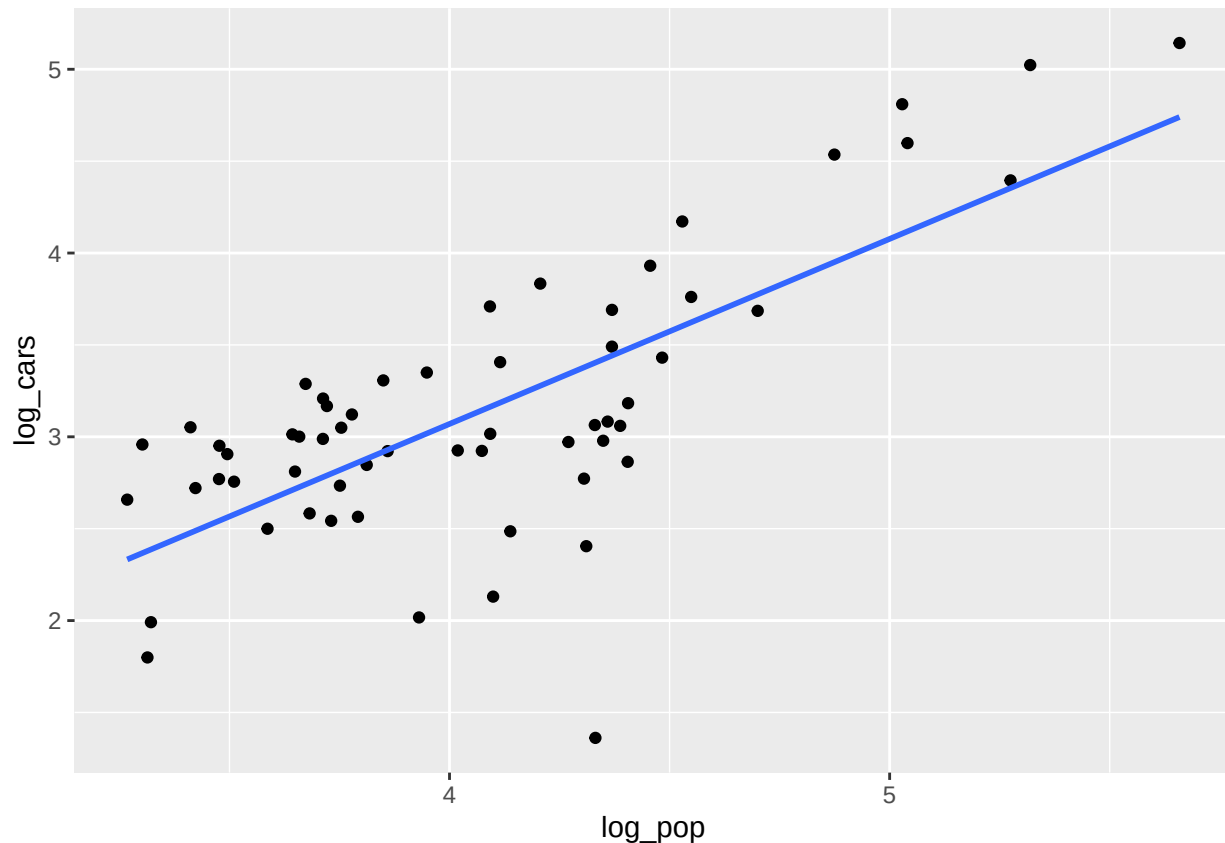
The relationship between the log_comp and log_pop is a strong, positive, linear relationship. If a line of best fit is drawn, it seems that the variability is constant throughout from the left to the right of the graph. As the log number of population increases, the log number of companies also increases.

4. Visualize one other variable with log population.

```
cit_log2<-cit %>%
  mutate(log_pop = log10(pop),
         log_cars = log10(cars))

ggplot(cit_log2, aes (x = log_pop, y = log_cars)) + geom_point() + stat_smooth(method = "lm", se = F)

## 'geom_smooth()' using formula 'y ~ x'
```



The relationship between the `log_cars` and `log_pop` is a moderate to strong, positive, linear relationship. If a line of best fit is drawn, the variability of `log_cars` seems to be larger when `log_pop` is smaller. This could be due to there's more cities with less smaller population, making it less data for the larger population side to show the variability. As the log number of population increase, the log number of care also increases.

5. Compute the correlation coefficient between log population and the log number of companies of these cities. Does it confirm your visual interpretation of their relation?

```
cor(cit_log$log_pop, cit_log$log_comp)
```

```
## [1] 0.8492927
```

```
m1 <- lm(log_pop ~ log_comp, data = cit_log)
coefficients(m1)
```

```
## (Intercept)    log_comp
##   2.5447799    0.6904834
```

The `r` value is 0.849, which means a strong correlation between the `log_pop` and `log_comp`. It does confirm the visual interpretation of their relationship. The slope is positive and linear, and the intercept is positive as well.

6. Compute the correlation between the log population and the log-transform of a variable of your choice (from question 4) to quantify the strength of the linear relation. Does the strength of the correlation surprise you?

```
cor(cit_log2$log_pop, cit_log2$log_cars)
```

```
## [1] 0.7257353
```

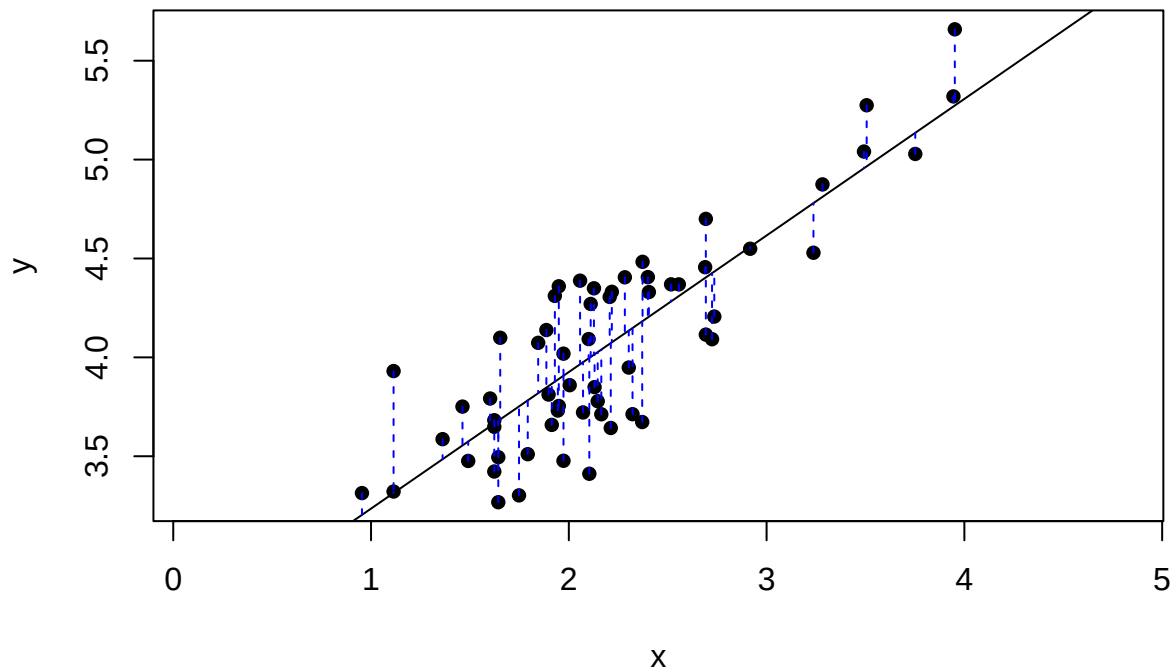
```
m2 <- lm(log_pop ~ log_cars, data = cit_log2)
coefficients(m2)
```

```
## (Intercept)    log_cars
##  2.4270782    0.5230964
```

The r value is 0.725, which means a moderate to strong correlation between the log_pop and log_cars. It does confirm the visual interpretation of their relationship. The slope is positive and linear, and the intercept is positive as well.

7. What are the coefficients of this linear model? Write out the form of the line that you could use to predict the log population of a city.

```
plot_ss(x = log_comp, y = log_pop, data = cit_log)
```



```
## Click two points to make a line.
## Call:
## lm(formula = y ~ x, data = pts)
##
## Coefficients:
## (Intercept)          x
##      2.5448      0.6905
##
## Sum of Squares:  4.703
```

The coefficient are a intercept of 2.5448, and a lsope of 0.6905. This, the line for predicting the log model is $\log_pop = 2.545 + 0.69 * \log_comp$.

8. Use the equation of the line you derived in the previous question to predict the value of log population for one new city. How does this predictions compare to the true value?

```
cit_new <- read_csv("https://www.dropbox.com/s/37cafpcj3lrenke/cit_new.csv?dl=1")
```

```
## Rows: 5 Columns: 15
```

```
## -- Column specification -----
## Delimiter: ","
## chr (2): city, state
## dbl (12): pop, pop_bra, pop_for, planted_area, area, gdp, gva, mun_exp, comp...
## lgl (1): hotels
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

cit_new <- cit_new %>%
  mutate(log_pop = log10(pop),
         log_comp = log10(comp),
         log_cars = log10(cars))

var2 <- tibble(log_comp = 1.342423)
predict(m1, var2)

##          1
## 3.471701
```

The true value for the log population for Francisco Dantas is 3.458487. The predicted value of 3.471701 seems to be really close to the value.

9. Using the original data set, compute a linear model for that predicts log population using a variable of your choice. Use that model to predict the value of log population for one of the new cities.

```
cit_log2 <- cit %>%
  mutate(log_pop = log10(pop),
         log_cars = log10(cars))

m2 <- lm(log_pop ~ log_cars, cit_log2)

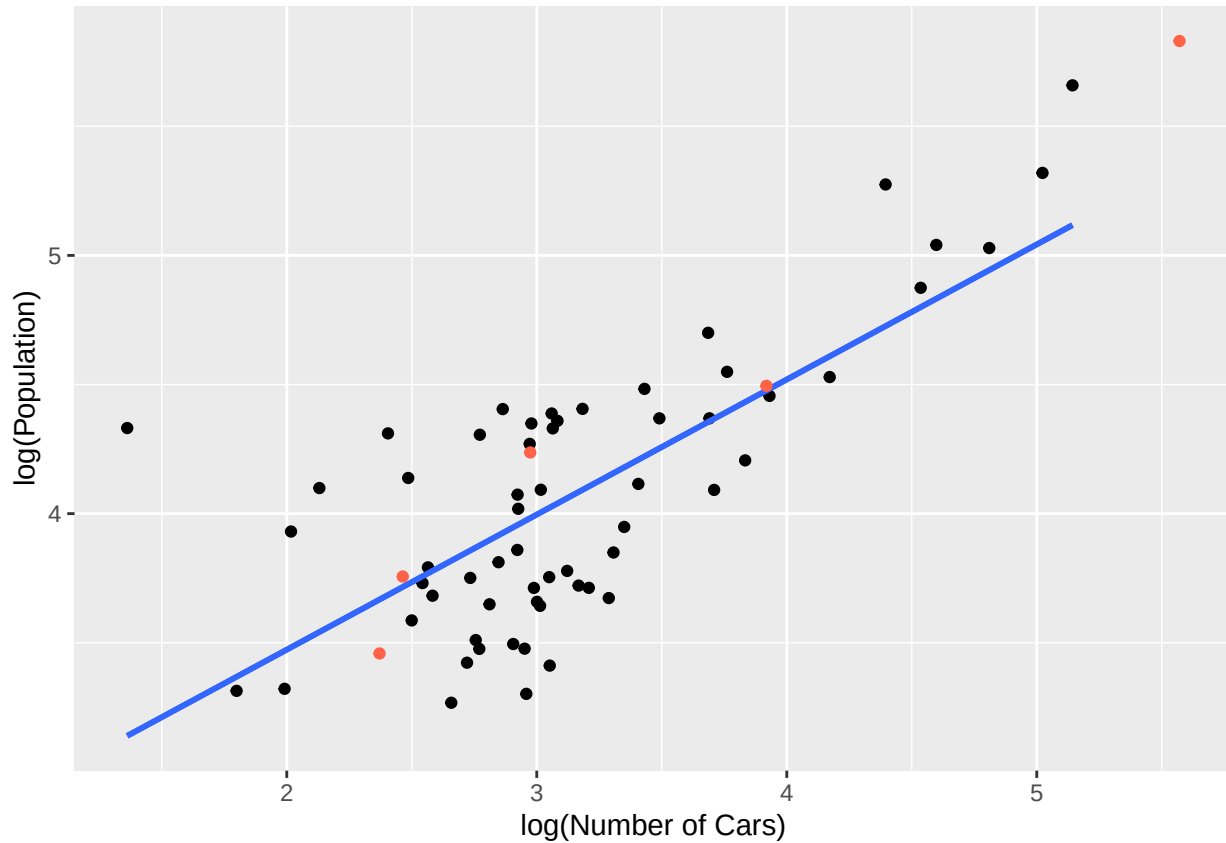
var <- tibble(log_cars = 2.371068)
predict(m2, var)

##          1
## 3.667375
```

10. Using your own variable of choice. Were the predictions that you model made for the 5 cities over or underestimates? Are the magnitudes of the errors generally smaller, larger, or the same as the residuals from the original data set? Do the predictions involve interpolation or extrapolation?

```
ggplot(cit_log2, aes(x=log_cars, y=log_pop)) +
  geom_point() + geom_smooth(method = "lm", se = FALSE) +
  stat_smooth(method = "lm", se = F) +
  labs(y = "log(Population)", x = "log(Number of Cars)") +
  geom_point(data = cit_new, aes(x=log_cars, y = log_pop), color = "tomato")

## 'geom_smooth()' using formula 'y ~ x'
## 'geom_smooth()' using formula 'y ~ x'
```



Four of the predictions are overestimate (2 of the four are pretty close to the estimation), while one is underestimate. The magnitudes of error generally are similar to the residuals in the original data set. The prediction involves interpolation as the predicted points in tomato are within the points/follows the trend within the original data set points. The new x value were not greater than the original x values Thus, we are predicting using values of x that are found within the range of x values to fit the model.